

Observability at TorIX: Custom Telemetry Solutions on Next-Gen Campus Switching



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Jeremy Foran
CEO, Blue Flag Consulting

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Data Engineer, Blue Flag Consulting

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Technical Marketing Engineer, Cisco
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2

Agenda

- 01 Introduction
- 02 Telemetry
- 03 Toronto Internet Exchange
- 04 Advanced Telemetry Solution
- 05 Resources

Cisco Webex App

Questions?

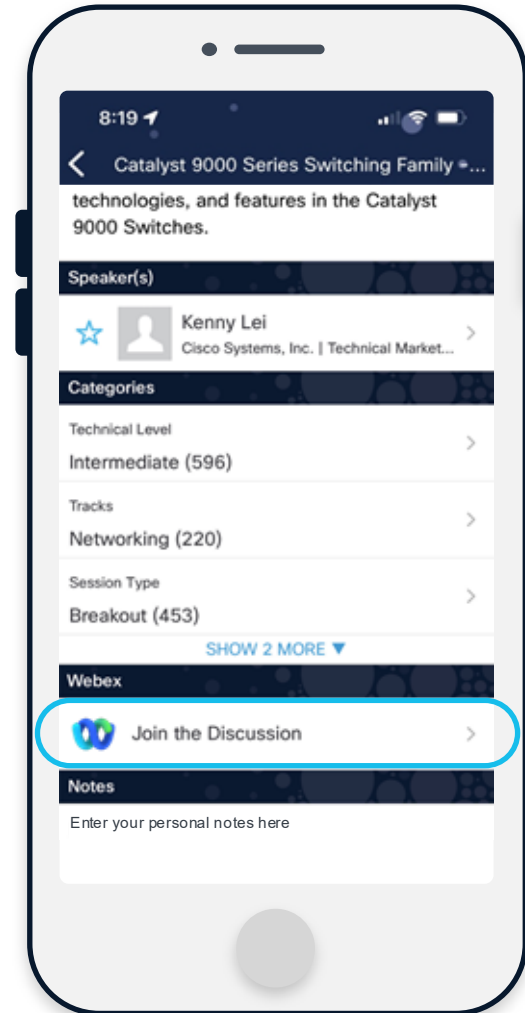
Use Cisco Webex App to chat with us!

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- 3 Install the Webex App or go directly to the Webex space
- 4 Enter messages/questions in the Webex space

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Who We Are

Jeremy Foran

CEO

Blue Flag Consulting



Yusuf Shalaby

Data Engineer

Blue Flag Consulting



Story DeWeese

Technical Marketing Engineer
Cisco

Intro to Telemetry

Programmable Interfaces

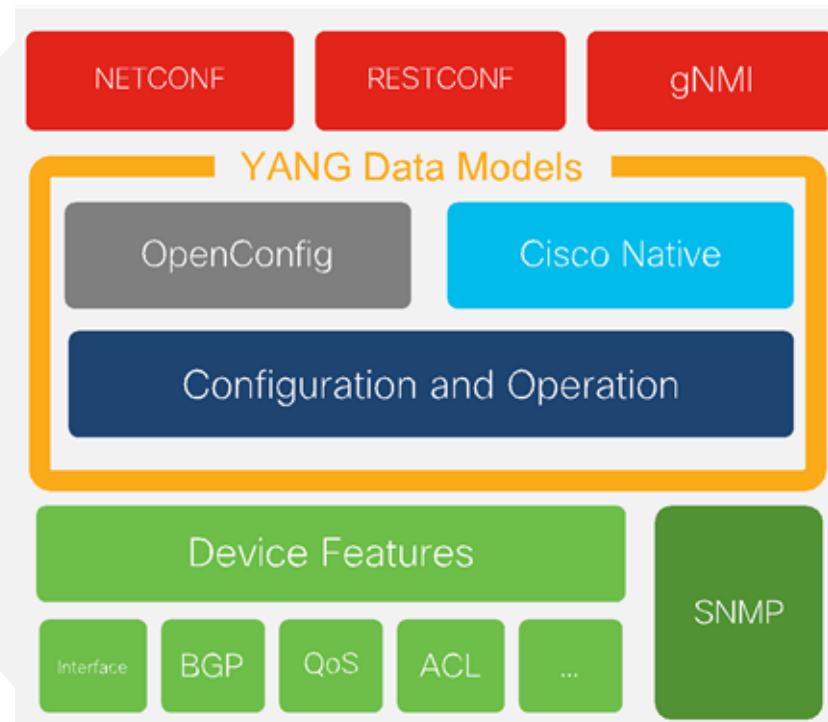
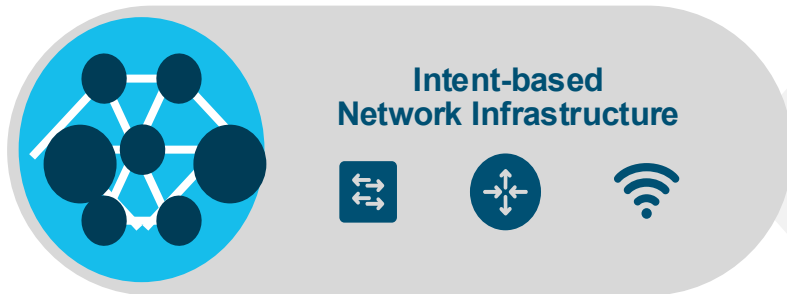
CLI

SNMP

WebUI

The NETCONF, RESTCONF and gNMI are programmatic interfaces that provide **additional** methods for interfacing with the IOS XE device – Just like the CLI, SNMP, and WebUI is used for configuration changes and operational metrics so can the programmatic interfaces of NETCONF, RESTCONF and gNMI

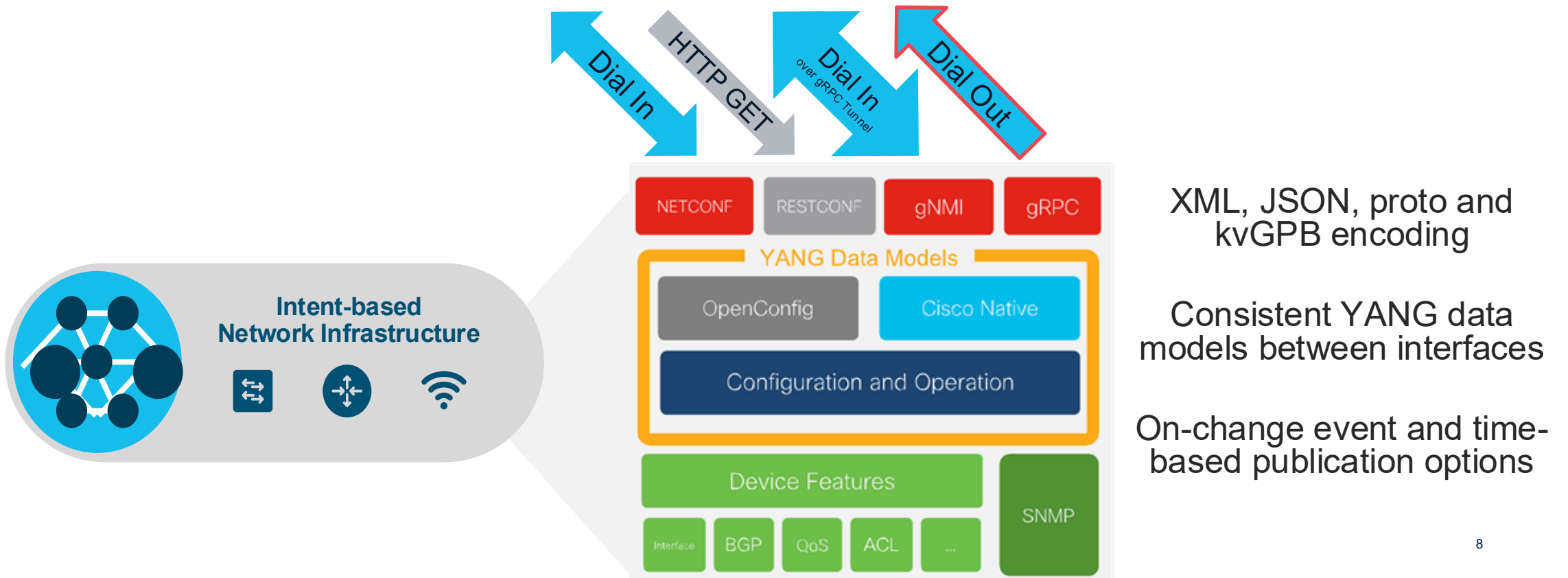
YANG data models define the data that is available for configuration and streaming telemetry



Model Driven Telemetry (MDT) Interfaces

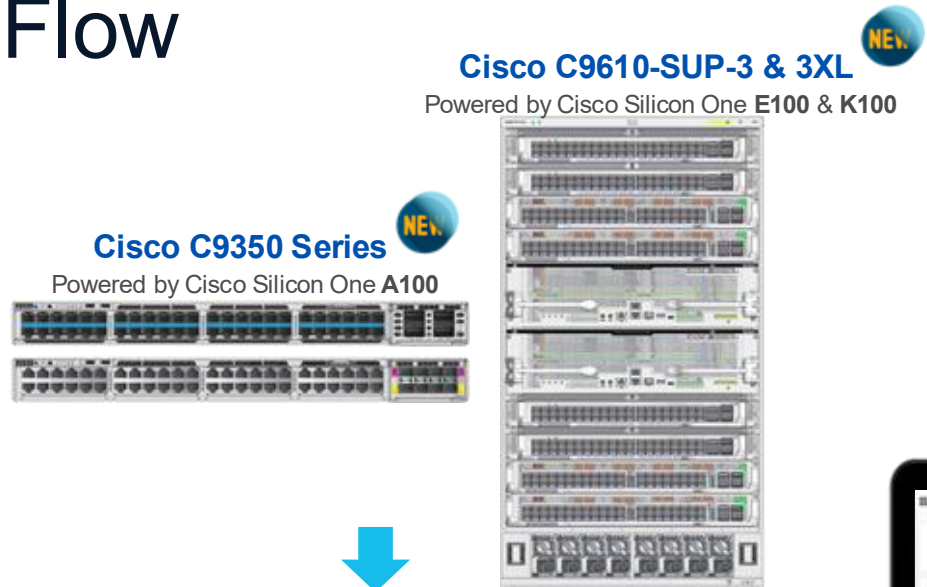
- ↔ Dial In: Collector establishes a connection to the device then subscribes to telemetry (pub/sub)
- ← Dial Out: Telemetry is pushed from the device to the collector based off configuration (push)

Publication / Subscription



Telemetry Data Flow

Cisco IOS XE Devices



Collector/Receiver
Decodes to text



Storage
Time Series Database



**Monitoring
and Visualizations**



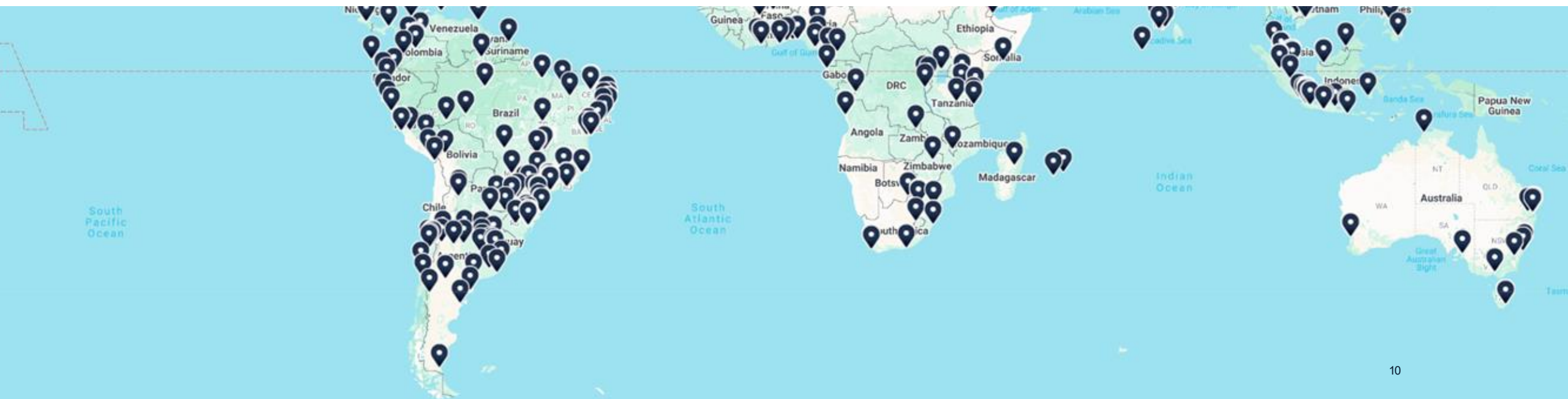
https://hub.docker.com/r/jeremycohoe/tig_mdt

<https://github.com/jeremycohoe/cisco-ios-xe-mdt>

https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/prog/configuration/1717/b_1717_programmability_cg/m_1717_prog_ietf_telemetry.html



INTERNET EXCHANGES





The Toronto Internet Exchange is
Canada's largest IXP, one of the largest
in the world.



2.3 tbps

peak traffic

270

organizations

312

networks

3rd

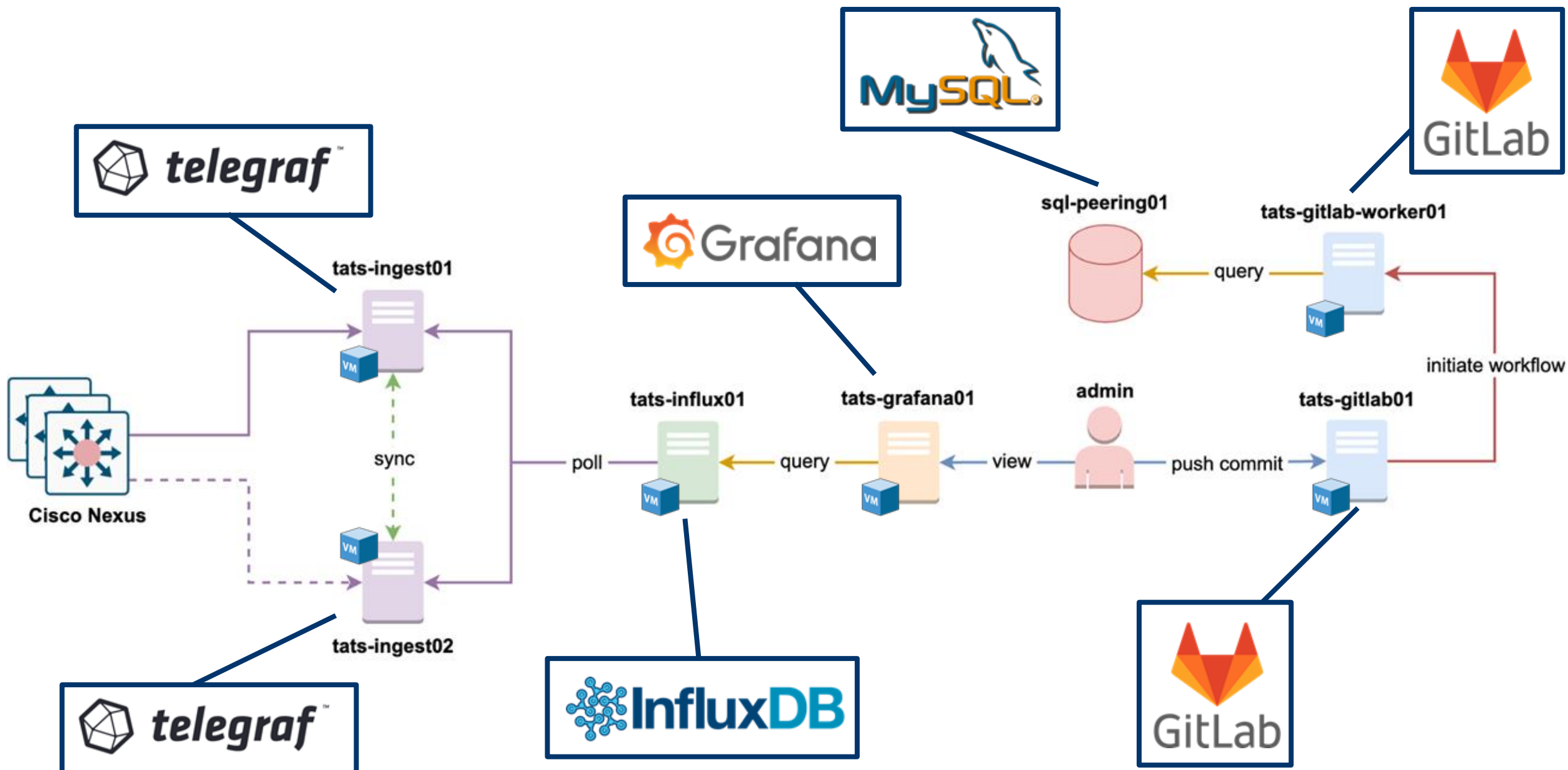
third largest NFPO in North America

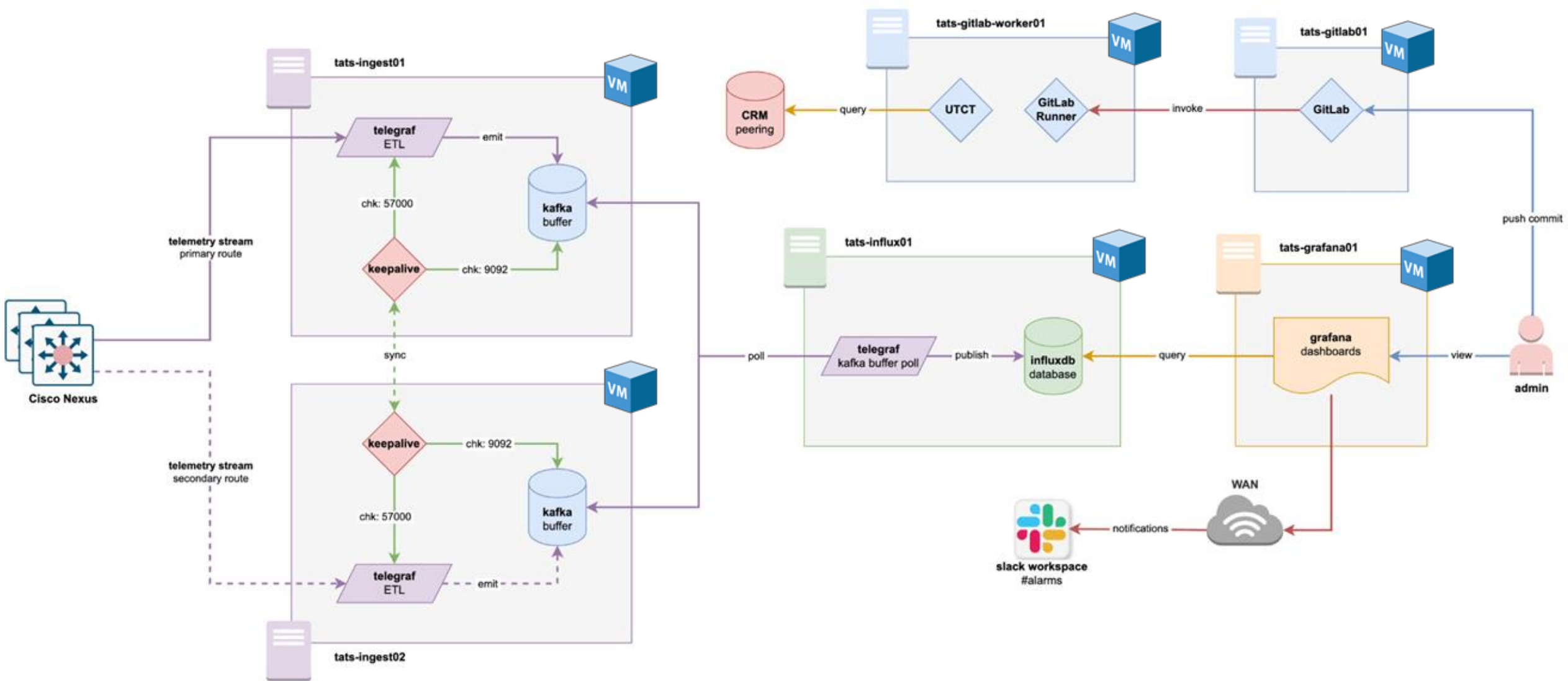


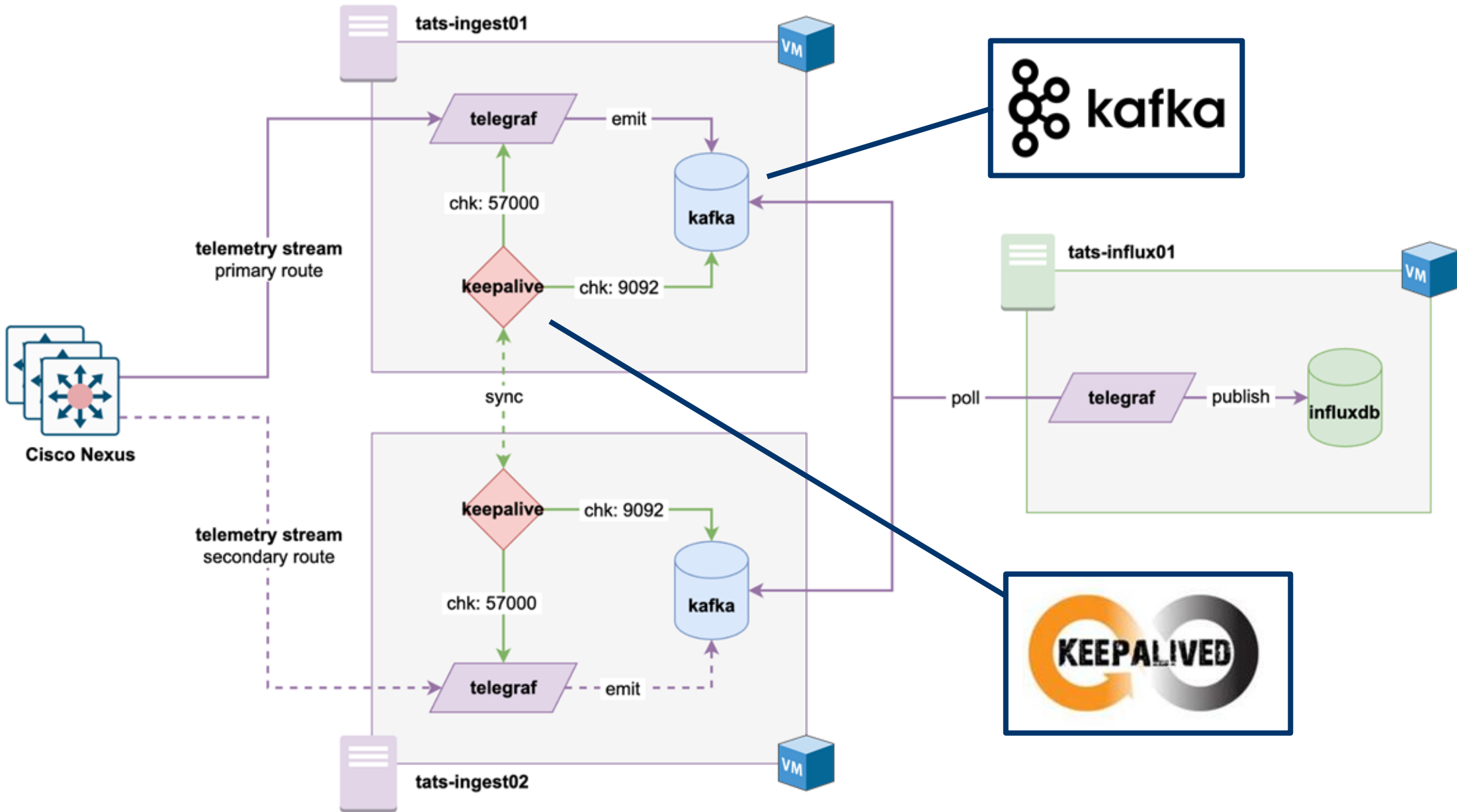
- **WHAT IS HAPPENING?**
- **WHAT HAPPENED?**
- **WHAT MIGHT HAPPEN?**
- **WHO IS AFFECTED?**
- **WHAT WAS THE ROOT CAUSE?**
- **WHEN SHOULD I TAKE ACTION?**

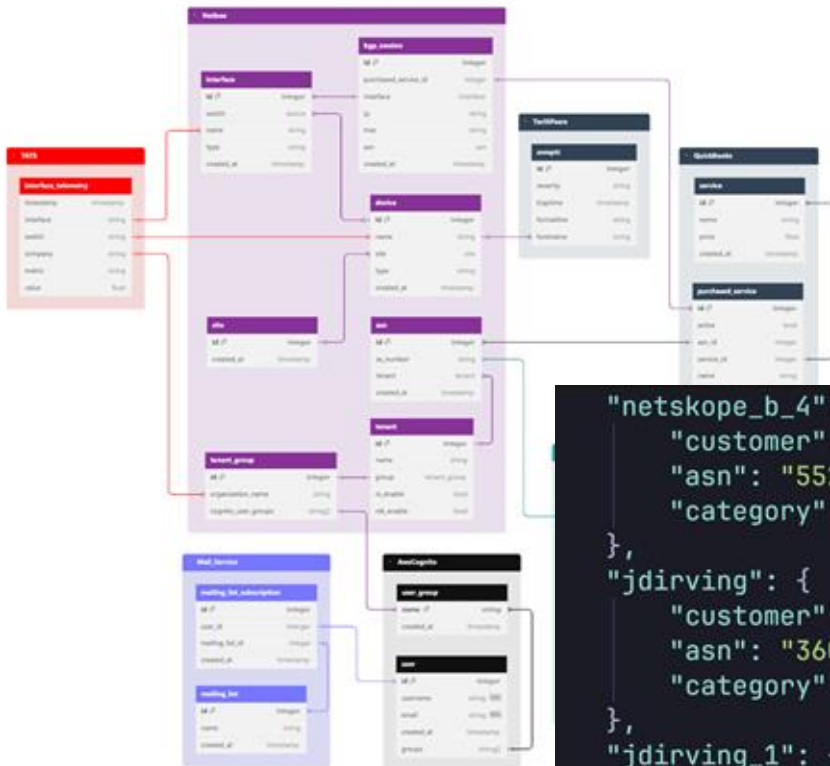
Advanced Telemetry Solution

Streamlined data. Structured for speed.









```

"netskope_b_4": {
  "customer": "Netskope, Inc.",
  "asn": "55256",
  "category": "Cloud"
},
"jdirving": {
  "customer": "J.D. Irving Inc.",
  "asn": "36004",
  "category": "Enterprise"
},
"jdirving_1": {
  "customer": "J.D. Irving Inc.",
  "asn": "36004",
  "category": "Enterprise"
},
"purplecow": {
  "customer": "Purple Cow Internet",
  "asn": "397545",
  "category": "ISP"
},
"purplecow_1": {
  "customer": "Purple Cow Internet",
  "asn": "397545",
  "category": "ISP"
},
"ssc_a": {
  "customer": "SSC",
  "asn": "2675",
  "category": "Enterprise"
}

```

```

root@torix-tats-runner-1: ~
root@torix-tats-runner-1:~# /usr/local/bin/torix-update-agent -configFile=/etc/torix-update-agent/conf

#####
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#
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# entity to whom it is addressed. Any dissemination, distribution, or
# unauthorized copying of this document is strictly prohibited.
#
#####

2025/06/10 23:38:11 INFO starting agent-updater version="build=[build=928297f buildTime=Thu Aug 22 08:
Jun 10 23:38:11.867 INF agent/main.go:31 loading configuration from file file=/etc/torix-update-agent/
Jun 10 23:38:11.867 INF torix/database.go:33 creating database client endpoint=10.135.25.120 port=3306
Jun 10 23:38:11.867 INF torix/database.go:70 pinging the database
Jun 10 23:38:11.869 INF torix/database.go:42 successfully connected to the database
Jun 10 23:38:11.869 INF torix/client.go:35 state file does not exist, creating a new state file file=s
Jun 10 23:38:11.870 INF agent/server.go:16 creating http server
Jun 10 23:38:11.870 INF agent/server.go:25 adding prometheus metrics endpoint /metrics
Jun 10 23:38:11.870 INF agent/server.go:29 adding updateCustomers endpoint /updateCustomers
Jun 10 23:38:11.870 INF agent/server.go:52 http server listening address=:8080
Jun 10 23:38:11.870 INF agent/daemon.go:35 entering main loop, listening for signals and events

```


Raw incoming data

```
sys/intf switch=602-es09-tor1,dn=sys/intf/phys-[eth1/31]/phys descr="purplecow" ... 1626357742000000000
sys/intf switch=602-es09-tor1,dn=sys/intf/phys-[eth1/31]/dbgIfIn octetRate=0,errors=0,discards=0, ... 1626357742000000000
sys/intf switch=602-es09-tor1,dn=sys/intf/phys-[eth1/31]/dbgIfOut octetRate=0,errors=0,discards=0, ... 1626357742000000000
```

After live join

```
sys/intf switch=602-es09-tor1,interface=eth1/31,descr="purplecow",dn=sys/intf/phys-[eth1/31]/dbgIfIn octetRate=0,errors=0,discards=0, ... 1626357742000000000
sys/intf switch=602-es09-tor1,interface=eth1/31,descr="purplecow",dn=sys/intf/phys-[eth1/31]/dbgIfOut octetRate=0,errors=0,discards=0, ... 1626357742000000000
```

```
"purplecow": {
  "customer": "Purple Cow Internet",
  "asn": "397545",
  "category": "ISP"
},
```

After lookups

```
sys/intf switch=602-es09-tor1,interface=eth1/31,descr="purplecow",dn=sys/intf/phys-[eth1/31]/dbgIfIn,asn=397545,category=ISP octetRate=0,errors=0,discards=0, ... 1626357742000000000
sys/intf switch=602-es09-tor1,interface=eth1/31,descr="purplecow",dn=sys/intf/phys-[eth1/31]/dbgIfOut,asn=397545,category=ISP octetRate=0,errors=0,discards=0, ... 1626357742000000000
```

Switch Cnt. ⓘ

22

Phys Port Cnt. ⓘ

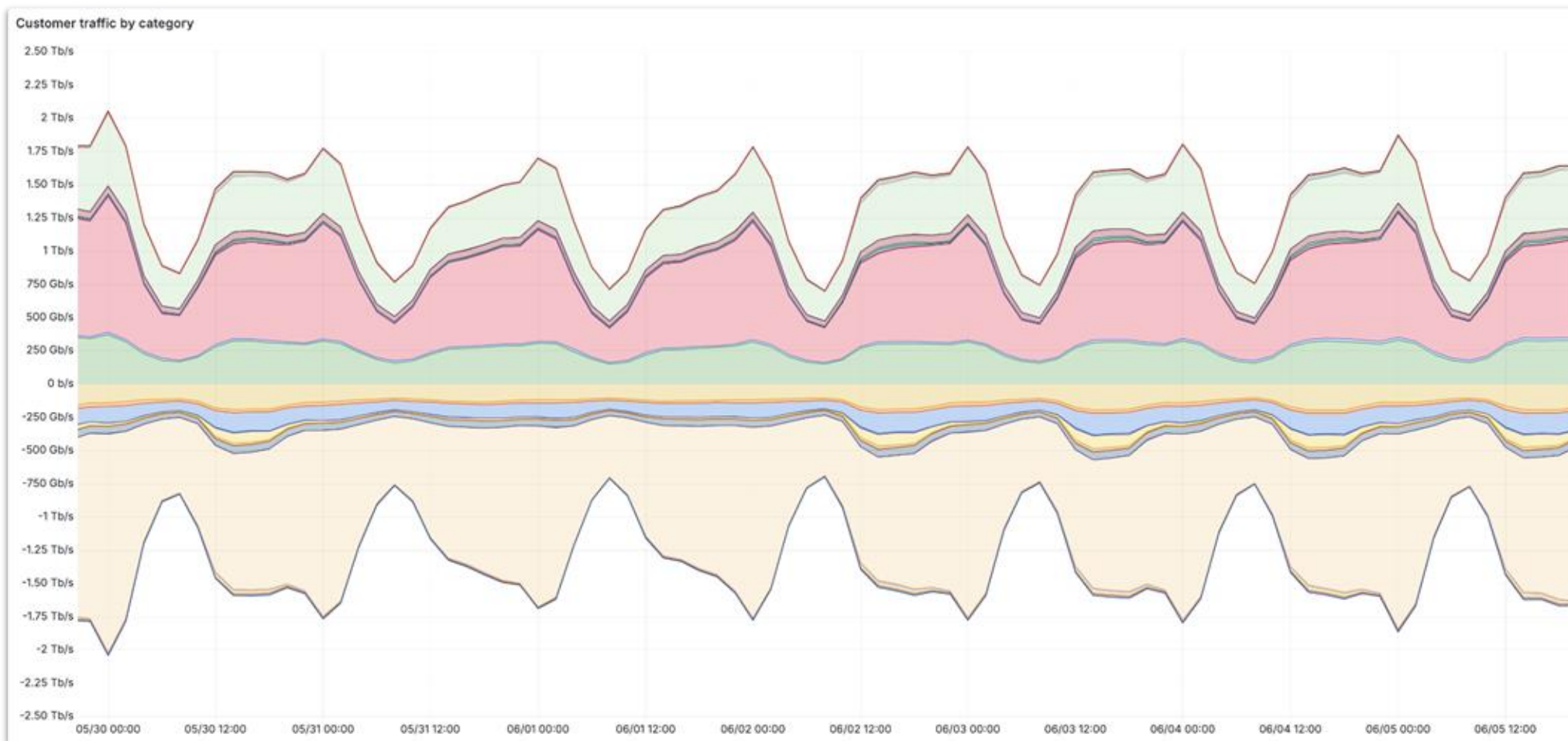
662

Aggr Port Cnt. ⓘ

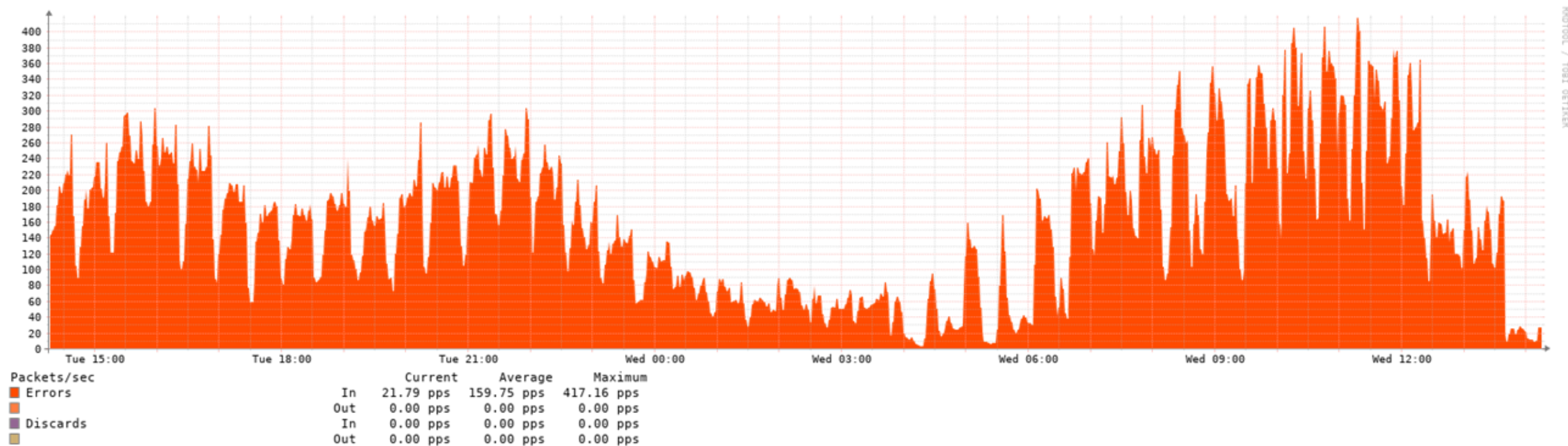
173

Customer Cnt. ⓘ

248

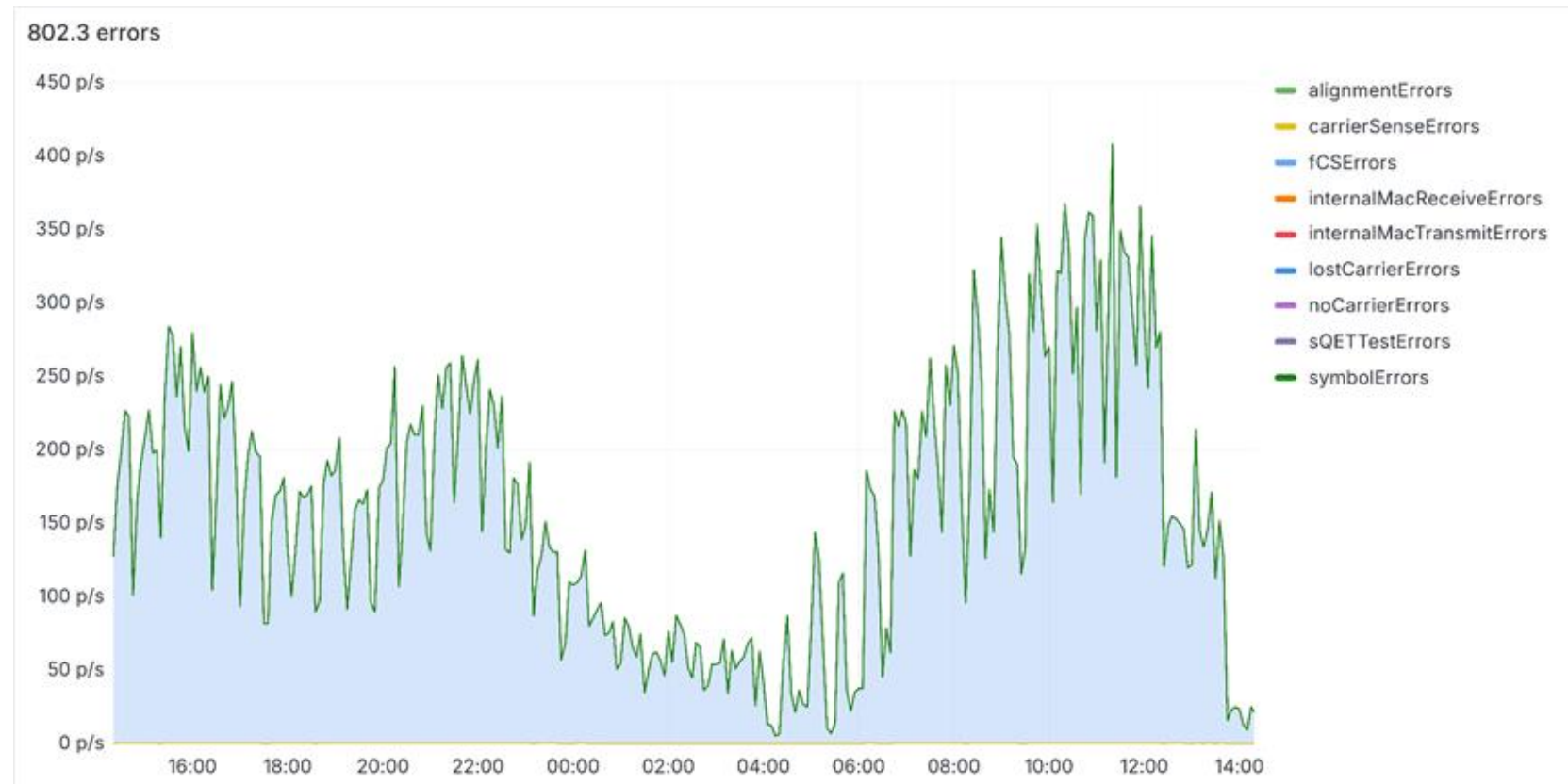


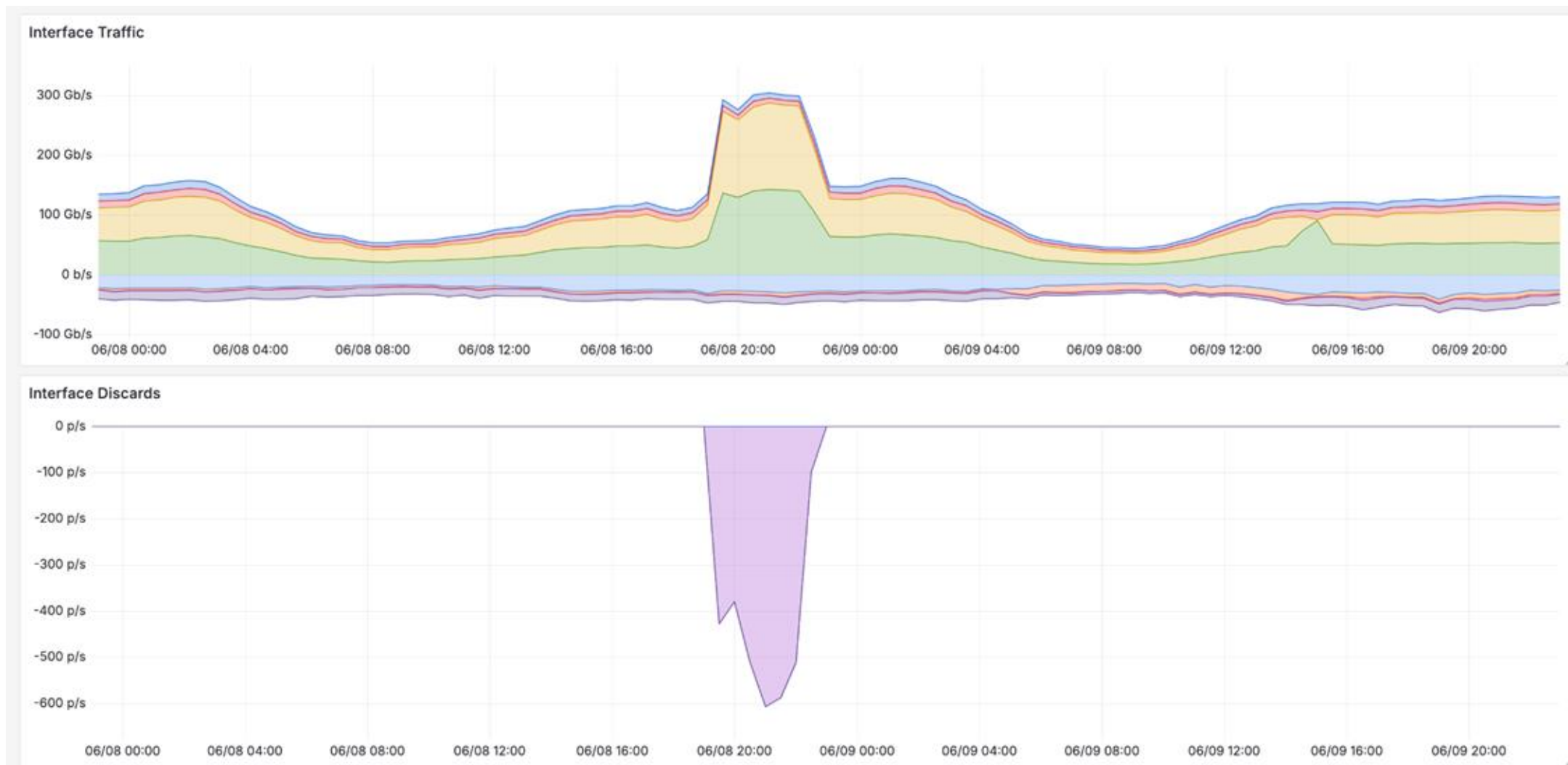
Errors on an interface - NMS



Errors on an interface - MDT + TIG

PROPERTY NAME	DATA TYPE	DESCRIPTION	POSSIBLE VALUES
alignmentErrors	scalar:UInt64	The alignment errors.	RANGE: [0, 18446744073709551615]
babble	scalar:UInt64	babble count	RANGE: [0, 18446744073709551615]
carrierSenseErrors	scalar:UInt64	The carrier sense errors.	RANGE: [0, 18446744073709551615]
controlInUnknownOpcodes	scalar:UInt64	The control input unknown opcodes.	RANGE: [0, 18446744073709551615]
deferredTransmissions	scalar:UInt64	The deferred transmissions.	RANGE: [0, 18446744073709551615]
excessiveCollisions	scalar:UInt64	The excessive collisions.	RANGE: [0, 18446744073709551615]
fCSErrors	scalar:UInt64	The frame check sequence errors.	RANGE: [0, 18446744073709551615]
frameTooLongs	scalar:UInt64	The frame too longs.	RANGE: [0, 18446744073709551615]
inPauseFrames	scalar:UInt64	The input pause frames.	RANGE: [0, 18446744073709551615]
inputDribble	scalar:UInt64	Input Dribble	RANGE: [0, 18446744073709551615]
internalMacReceiveErrors	scalar:UInt64	The internal MAC receive errors.	RANGE: [0, 18446744073709551615]
internalMacTransmitErrors	scalar:UInt64	The internal MAC transmit errors.	RANGE: [0, 18446744073709551615]
lateCollisions	scalar:UInt64	The late collisions.	RANGE: [0, 18446744073709551615]
lostCarrierErrors	scalar:UInt64	Lost Carrier Errors	RANGE: [0, 18446744073709551615]
multipleCollisionFrames	scalar:UInt64	The multiple collision frames.	RANGE: [0, 18446744073709551615]
noCarrierErrors	scalar:UInt64	No Carrier Errors	RANGE: [0, 18446744073709551615]
outPauseFrames	scalar:UInt64	The output pause frames.	RANGE: [0, 18446744073709551615]
runts	scalar:UInt64	runts count	RANGE: [0, 18446744073709551615]
sQETTestErrors	scalar:UInt64	The SQET test errors.	RANGE: [0, 18446744073709551615]
singleCollisionFrames	scalar:UInt64	The single collision frames.	RANGE: [0, 18446744073709551615]
symbolErrors	scalar:UInt64	The symbol errors.	RANGE: [0, 18446744073709551615]

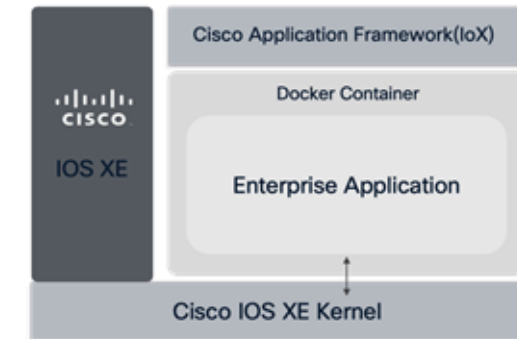




BLUE FLAG Ingestion at the Edge

What is Application Hosting?

Cisco Application Hosting on Catalyst 9000 series of switches extends the cloud to the edge of the network enabling data processing closer to the source in a distributed manner. Application Hosting is used by businesses ranging from manufacturing and energy corporations to public sector organizations to host new breed for Security, network monitoring/troubleshooting and data analytics tools while removing the need for expensive compute nodes, as application hosted on the switch gives developer easy and efficient access to live network traffic as well as the rich telemetry supported by the infrastructure. Application Hosting allows to run off the shelf **Docker apps** it helps shortens development cycle.

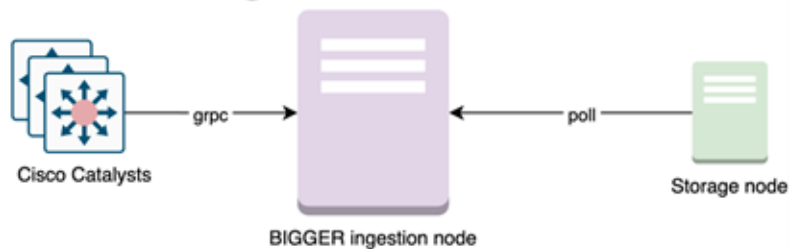


Cisco C9350 Series **NEW**
Powered by Cisco Silicon One **A100**

<https://www.cisco.com/c/en/us/products/collateral/switches/catalyst-9300-series-switches/white-paper-c11-742415.html>

BLUE FLAG Ingestion at the Edge

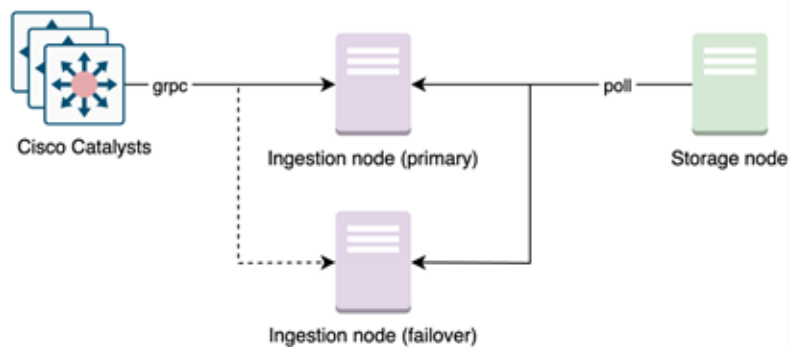
Vertical scaling



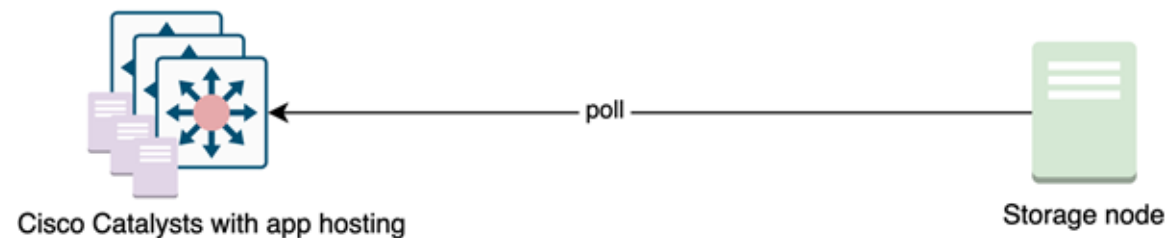
Horizontal scaling with load balancer



Failover with virtual IP



Edge processing and caching





Resources

Cisco University (Cisco U) part of L&D

<https://u.cisco.com>

<https://u.cisco.com/search/tutorial?query=Story%20DeWeese,%20Jeremy%20Coho,%20not%20berry>

Direct link to Tutorial, requires login to u.cisco.com first:

1. <https://ondemandlearning.cisco.com/apollo-alpha/tc-iosxe-ztp/pages/1>
2. <https://ondemandlearning.cisco.com/apollo-alpha/tc-terraform-ios-xe/pages/1>
3. <https://ondemandlearning.cisco.com/apollo-alpha/tc-yangsuite-netconf/pages/1>
4. <https://ondemandlearning.cisco.com/apollo-alpha/tc-yangsuite-restconf/pages/1>

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TUTORIAL

 Jeremy Cohoe

Cisco IOS XE Zero-Touch Provisioning

In this tutorial, you will learn the basics of the IOS XE ZTP feature.

Beginner 45m

Created October 23, 2023

TUTORIAL

 Story DeWeese

Using YANG Suite with NETCONF

In this tutorial, you'll learn how to use YANG Suite to visualize the NETCONF protocol with network devices.

Intermediate 30m

Created October 24, 2023

TUTORIAL

 Story DeWeese

Using YANG Suite with RESTCONF and gRPC

In this tutorial, you'll learn how to use YANG Suite to visualize the RESTCONF and gRPC protocols with network devices.

Intermediate 10m

Created October 24, 2023

TUTORIAL

 Story DeWeese

Cisco IOS XE Terraform Provider

Get started leveraging the power of Terraform. In this tutorial, we'll use this cloud-native provisioning tool to create a new VLAN.

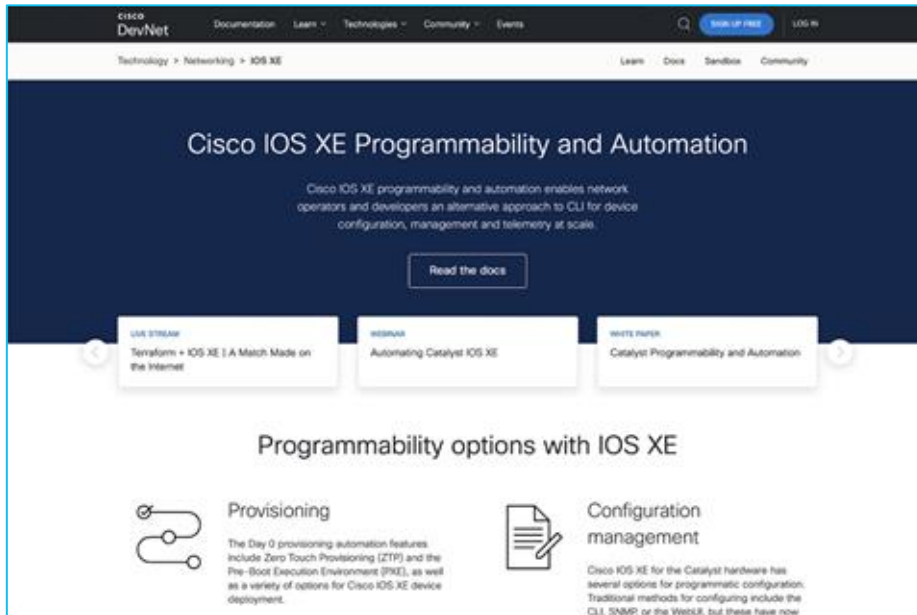
 

Beginner 15m

Created February 21, 2023

Programmability Website

The one-stop-shop for Cisco IOS XE Programmability resources including videos, white papers, labs and more!



- Community Forum
- IOS XE FAQ
- White Papers
- Code Exchange
- IOS XE Docs & Guide
- Learning Tracks and Labs
- Sandboxes
 - ... and more !



<https://developer.cisco.com/iosxe/>

Cisco YANG Suite



YANG API Testing and Validation Environment

Construct and test YANG based APIs over
NETCONF, RESTCONF, gRPC and gNMI

IOS XE / IOS XR / NX OS platforms

Get hands-on using the new learning lab!

<https://developer.cisco.com/learning/labs/intro-yangsuite/>

Docker container innovation 1 container

The screenshot displays the Cisco YANG Suite web interface. The top section, titled 'Explore YANG Models', shows a tree of nodes for the 'Cisco-IOS-XE-interfaces-oper' module. The bottom section, titled 'NETCONF', shows a 'YANG Tree' and a 'Value' field. The right side displays 'Node Properties' for the selected node, including Name, Nodetype, Description, Module, Revision, Xpath, Prefix, and Namespace.

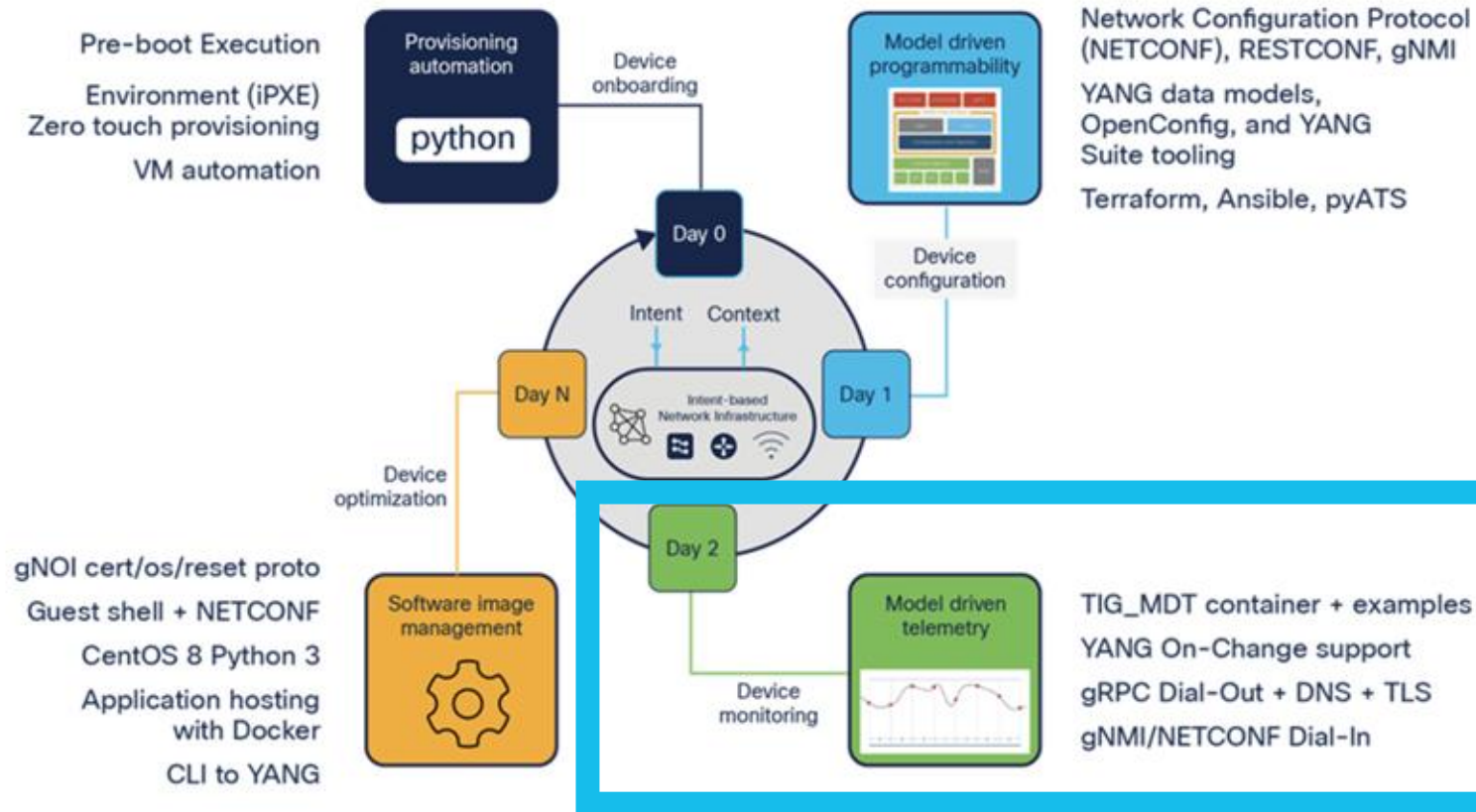
developer.cisco.com/yangsuite

github.com/CiscoDevNet/yangsuite



Ready to learn more?

White Paper on this telemetry use case coming soon!



Deeper dive into this telemetry use case!

API White Paper

Programmability and auto... ^ Q

Table of Contents

Programmability and automatio... -

Day 0: Provisioning automation

Day 1: Model-driven programmability

Day 2: Model-driven telemetry

Day N: Device optimization

Cisco IOS XE operational consistency

Yet Another Next Generation (Y... +

Day 1: Model-driven program... +

Tooling: Cisco YANG Suite +

Day 2: Model-driven telemetry +

Day N: Device optimization +

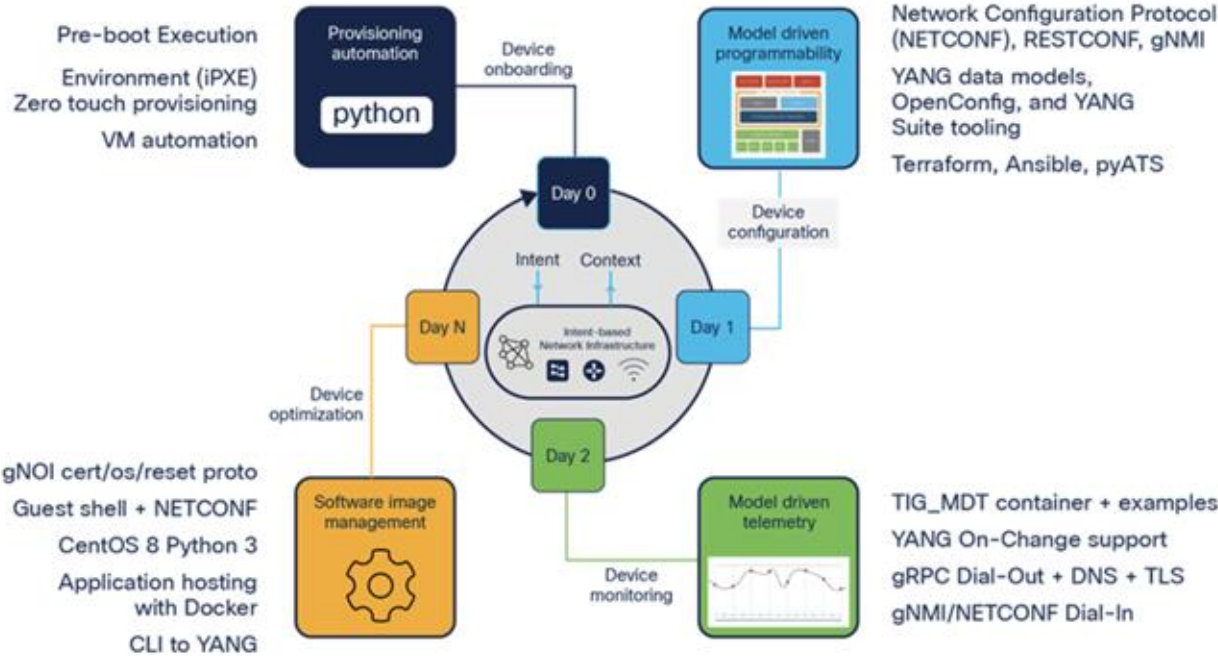
Conclusion

Additional resources +

Blogs

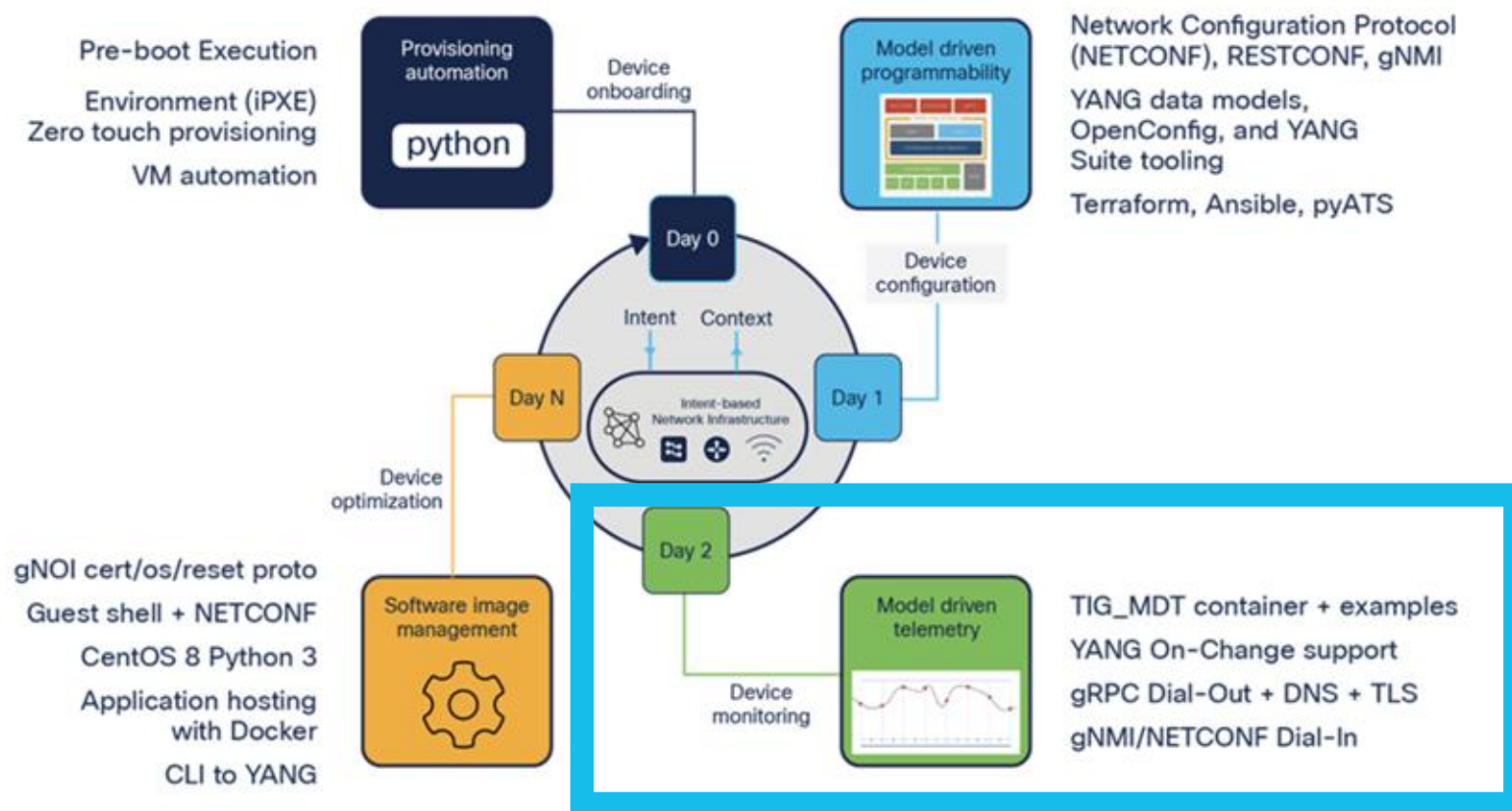
Products & Services / Switches / Campus LAN Switches - Access / Cisco Catalyst 9300 Series Switches /

Catalyst Programmability and Automation



Model Drive Telemetry (MDT) White Paper

The Model Driven Telemetry White Paper includes examples, use cases and tooling related to telemetry.
This paper is now available online and in PDF form!



View online: <https://www.cisco.com/c/en/us/products/collateral/switches/catalyst-9300-series-switches/model-driven-telemetry-wp.html>
View as PDF: <https://www.cisco.com/c/en/us/products/collateral/switches/catalyst-9300-series-switches/model-driven-telemetry-wp.pdf>

Table of Contents

Introduction to Cisco IOS XE

Introduction to telemetry

Benefits of model driven tele...

Network monitoring challeng...

Architecture and databases

Dial-in and dial-out MDT +

Publication notification optio...

YANG data modeling language +

Benchmarking and comparis... +

Cisco controller solutions +

Cloud solutions +

Tooling +

Dashboarding and validation +

Configuration examples +

Telemetry configuration man... +

Troubleshooting and validati... +

Best practices and lessons l... +

Conclusion

Resources

dCloud Programmability

<https://dcloud.cisco.com>

“Cisco Catalyst 9000 IOS XE Programmability & Automation Lab v1”

<https://dcloud2.cisco.com/demo/catalyst-9000-ios-xe-programmability-automation-lab-v1>

Use Cases:

EVPN:

- Ansible with CLI deployment of EVPN solutions
- EVPN management over RESTCONF/YANG with Postman
- Declarative EVPN fabric management with Terraform

Tooling and Integrations

YANG Suite

- NETCONF/RESTCONF/gNMI API
 - Ansible integration
- NETCONF/gNMI Dial-In Telemetry
- gRPC Dial-Out Telemetry receiver

Telemetry

- TIG stack in Docker
- Grafana dashboard for device health

Postman / RESTCONF

- EVPN fabric API calls

Terraform/RESTCONF

- Declarative EVPN fabric management

Ansible

- EVPN solution enablement using CLI

Model Driven Telemetry

Telemetry configuration with CLI and YANG Suite

Collection with TIG_MDT container and tooling

YANG Programmability

YANG Suite tooling and integrations to YANG API's

Ansible integrations

Ubuntu VM Details:

Syslog receiver from all switches
TFTP config backup
See slide

Windows VM Details

VS Code

Terraform @ folder
Ansible @ folder

Chrome browser

YANG Suite, Grafana

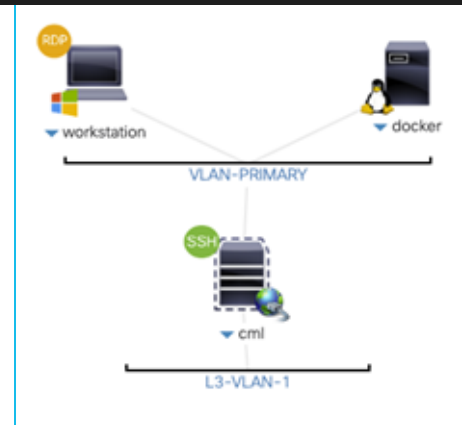
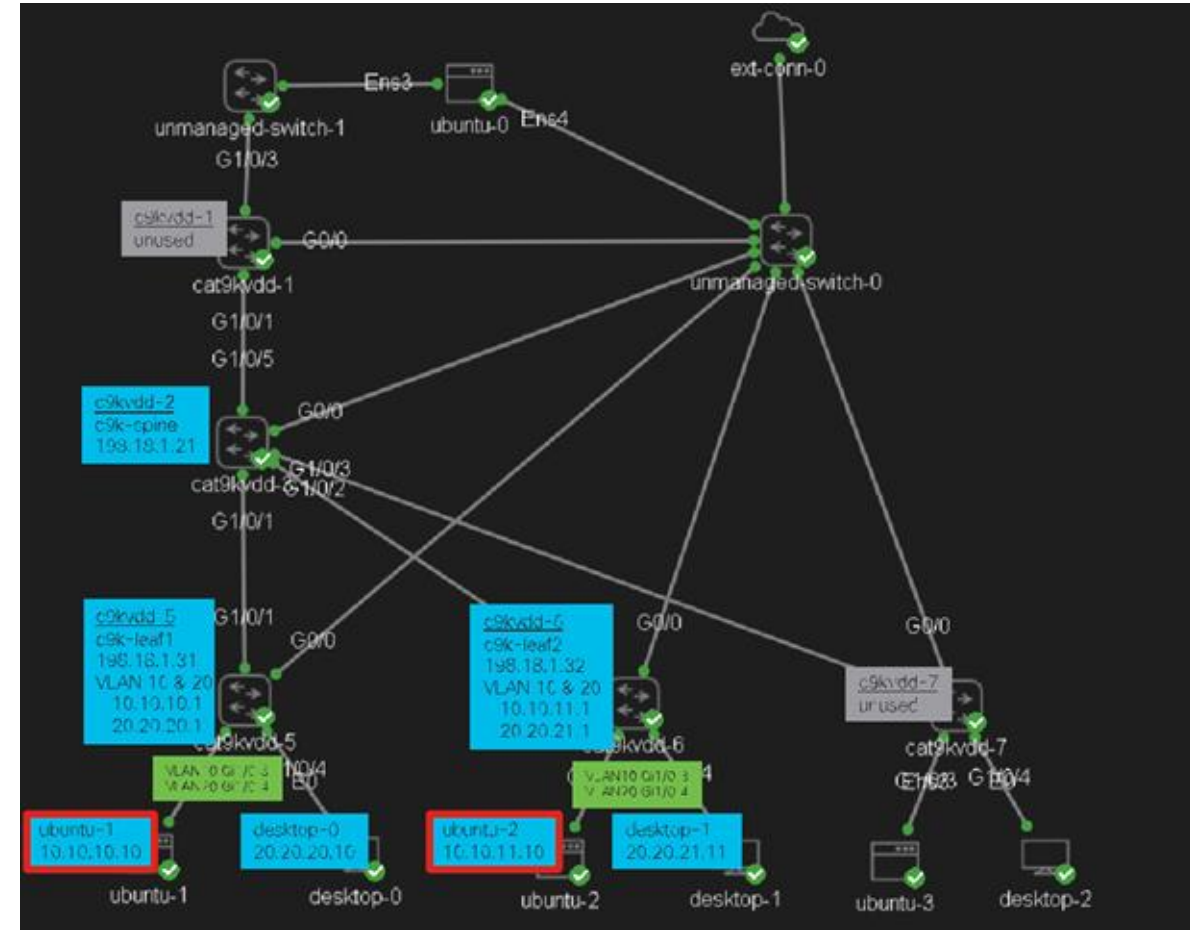
Bash/PS/Cmd shells

SSH into C9K or Ubuntu

Postman

Workspace for EVPN

C9K VM's



VLAN1	
c9k-spine	IP: 198.18.1.21 developer /
C1sco12345	
c9k-leaf1	IP: 198.18.1.31 developer /
C1sco12345	
c9k-leaf2	IP: 198.18.1.32 developer /
C1sco12345	
c9kvdd-1 - unconfigured	

c9kvdd-7 - unconfigured

Cisco IOS XE - Reservable Virtual Sandbox

This DevNet reservable sandbox has IOS XRv + N9Kv + IOS XE
The Catalyst 8000V within the DevBox is enabled for Day 0/ZTP usecases

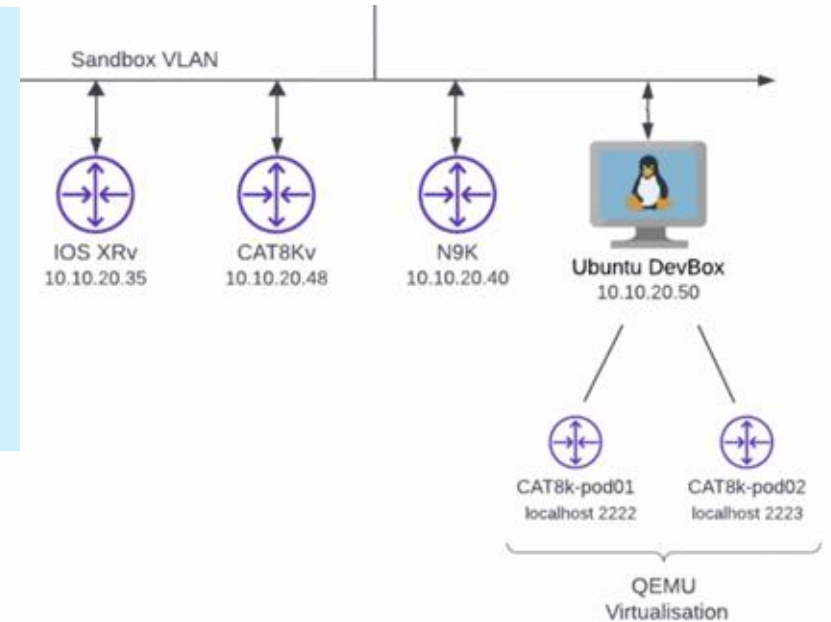
<https://devnetsandbox.cisco.com>

“IOS XE on Cat8kv”

<https://devnetsandbox.cisco.com/DevNet/catalog/IOS%20XE%20on%20Cat8kv>

The “Enterprise Networking” Learning Labs contains guides for the supported usecases

<https://developer.cisco.com/learning/>



Explore YANG data models and NETCONF to query and configure IOS XE network devices. Then use Ansible to manage configurations on your IOS XE devices, covering the fundamentals of Ansible. Lastly, enable...

🕒 6 hrs - 4 modules - 12 labs

IOS XE on Cat8kv

The IOS XE on Catalyst 17.9.2 Sandbox offers developers access to an IOS XE device running the latest IOS XE code release available on cisco.com. Here you can test out the newest programmability...

Networking

reservable



Launch

Sandbox Capabilities:

TIG_MDT Telemetry
YANG Suite API
Terraform + Ansible
ZTP & Guest Shell
... and more

Table of Contents

- Overview
- Network Topology
- Grafana Dashboard
- Access Details
- Additional Resources
- VPN Access

Catalyst 8000v Credentials

host	port	username	password
10.10.20.48	22	developer	C1sco12345
10.10.20.48	830	developer	C1sco12345
10.10.20.48	443	developer	C1sco12345

Developer Box Credentials

host	port	username	password
10.10.20.50	22	developer	C1sco12345
telnet localhost (CAT8k-1)	2222	developer	C1sco12345
telnet localhost (CAT8k-2)	2223	developer	C1sco12345

Cisco IOS XE - Always On Virtual Sandbox

The “IOS XE on Cat8kv Always On” virtual sandbox requires no reservation

The c8kv can be accessed with SSH, NETCONF, and RESTCONF

Hostname: sandbox-iosxe-latest-1.cisco.com

Username: admin

Password: C1sco12345

SSH port 22, NETCONF port 830, RESTCONF HTTPS

<https://devnetsandbox.cisco.com>

devnetsandboxiosxe.cisco.com

<https://devnetsandbox.cisco.com/DevNet/catalog/ios-xe-always-on>

The “Enterprise Networking” Learning Labs contains guides for the supported usecases

<https://developer.cisco.com/learning/>

IOS XE on Cat8kv AlwaysOn



This AlwaysOn sandbox offers developers access to an IOS XE device running the latest IOS XE code release available on cisco.com (currently 17.11.x). Here you can test out the newest programmability feature...

Always-On



Launch

Access Details:

Developers and network engineers access the IOS XE on Catalyst 17.11.x sandbox directly using the following information:

- Cat8000v Host
 - Address: `sandbox-iosxe-latest-1.cisco.com`
 - Username: admin
 - Password: C1sco12345
 - NETCONF port: 830
 - gRPC telemetry port: 57500
 - ssh port: 22

Learning Track

Enterprise Networking

Explore YANG data models and NETCONF to query and configure IOS XE network devices. Then use Ansible to manage configurations on your IOS XE devices, covering the fundamentals of Ansible. Lastly, enable...

6 hrs - 4 modules - 12 labs

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Questions in the back!

Thank you

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